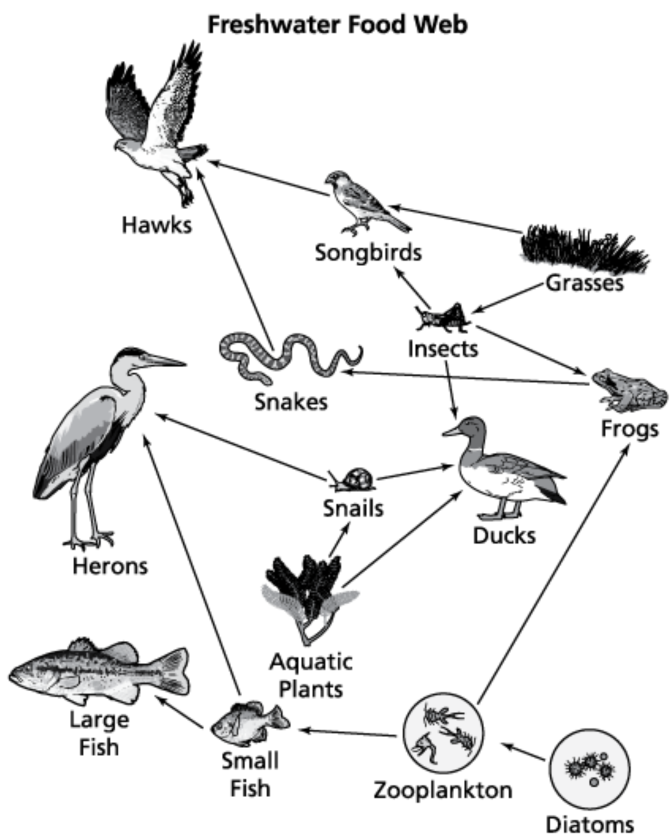
Tennessee Comprehensive Assessment Program

# TCAP

## Science
## Grade 6
## Practice Items


TN Department of Education

The diagram shows a food web in a freshwater ecosystem.




Which **three** food chains correctly represent paths of energy flow through this ecosystem?

M. Small Fish → Zooplankton → Diatoms

P. Aquatic Plants → Snails → Herons

R. Zooplankton → Frogs → Snakes

S. Insects → Songbirds → Hawks

T. Small Fish → Large Fish → Herons

Hydrilla is a plant that grows in freshwater. This plant is not native to Tennessee but is found in many of Tennessee’s rivers. Hydrilla has become a problem because it crowds out native plants. One way to reduce the amount of hydrilla is to release a species of nonreproductive carp that will eat the hydrilla and other freshwater plants.

Which step should be taken before releasing carp into the rivers to reduce the amount of hydrilla?

A. Interview residents about how hydrilla has affected people living near rivers.

B. Provide demonstrations on fishing for carp in rivers.

C. Determine how the carp will impact native Tennessee water plants.

D. Educate people about the carp and their habits.

The data table lists estimates of the percentage of energy consumed from different sources in one area in the United States in 1815, 1915, and 2015.

**Estimates of Energy Consumption by Source**

| | Energy Source | | | | | | |
|---|---|---|---|---|---|---|---|
| **Year** | **Wood** | **Coal** | **Petroleum** | **Natural Gas** | **Hydroelectric** | **Nuclear** | **Other** |
| 1815 | 100% | 0% | 0% | 0% | 0% | 0% | 0% |
| 1915 | 10% | 75% | 5% | 5% | 0% | 0% | 0% |
| 2015 | 2% | 23% | 35% | 20% | 0% | 10% | 5% |

Which statement explains the **most likely** reason that sources of energy used by people in this area changed over time?

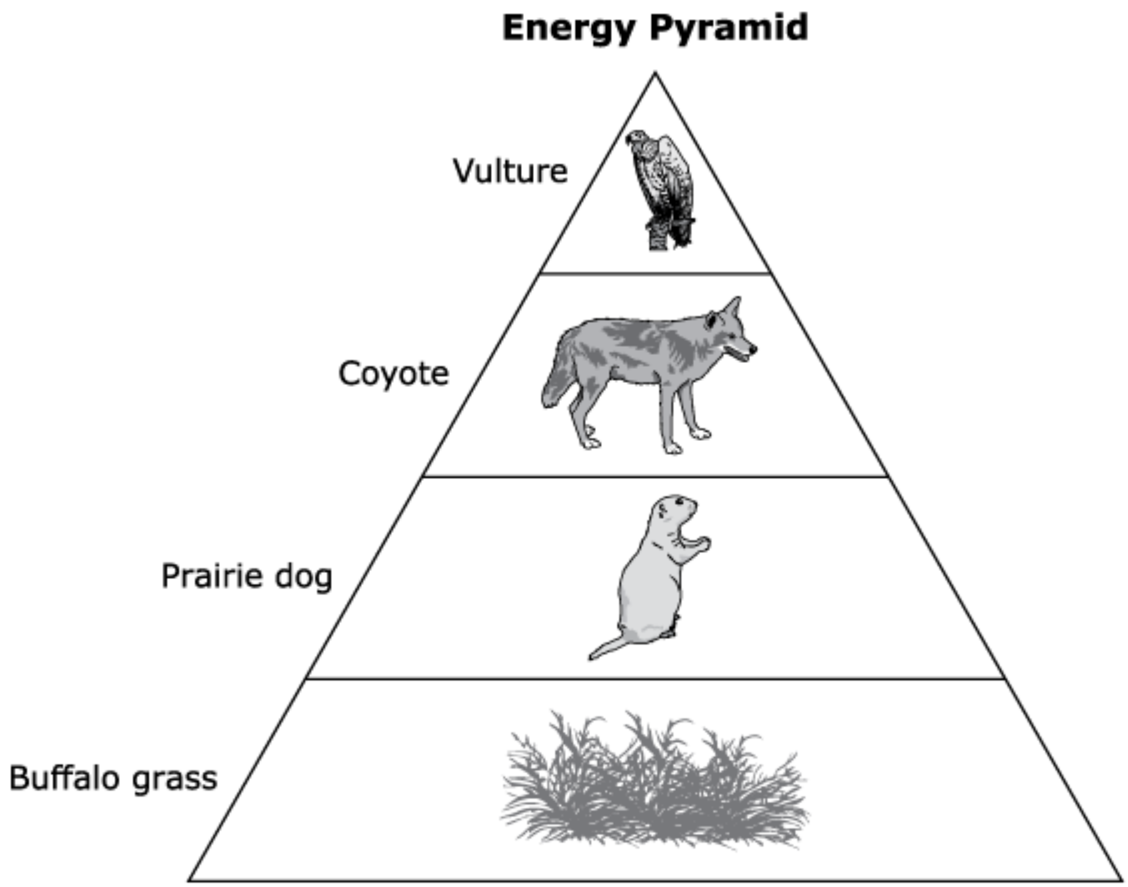
M. Engineers have invented new ways of getting energy resources from the earth.

P. No more wood is available to be transformed into energy in the United States.

R. Modern devices, such as mobile phones and computers, can only use energy from nuclear sources.

S. Hydroelectric sources have proven difficult to transform into useful forms of energy.

The figure shows an energy pyramid for some organisms in a grassland ecosystem.




Which statement **best** describes energy changes between levels of this energy pyramid?

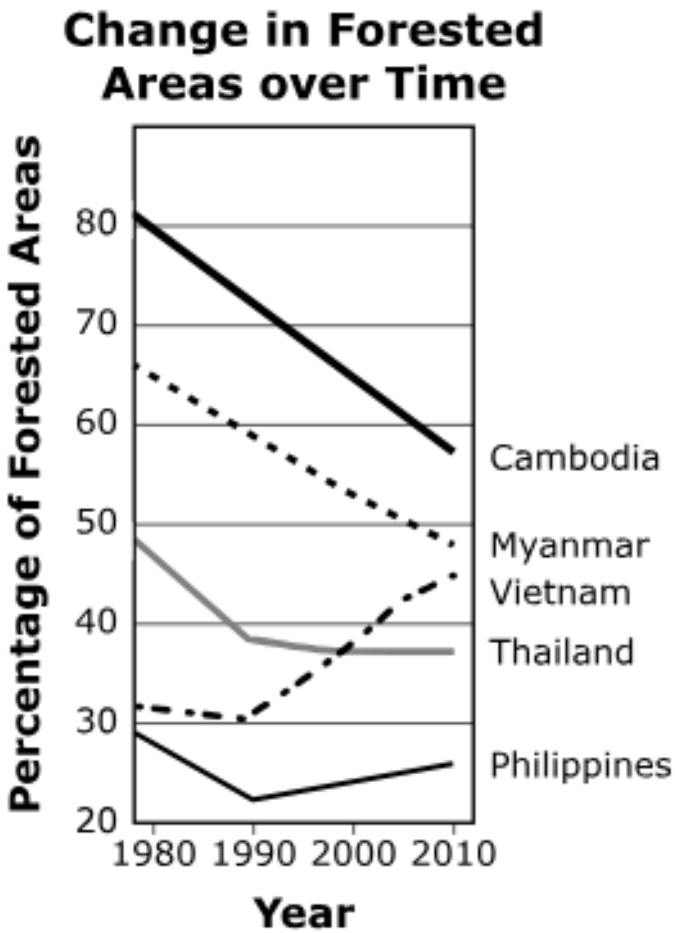
A. The energy transfers from the vulture to the coyote.

B. The energy increases from the buffalo grass to the prairie dog.

C. 90% of the available energy in one trophic level will pass on to the next.

D. 10% of the available energy in one trophic level will pass on to the next.

As humans harvest forested areas, they often reinvest in preserving these areas through reforestation or sustainability plans. The figure shows the percentage of forested areas in several countries.




Using the information provided in the graph, which **two** countries have **most likely** not developed a reforestation plan?

M. Cambodia, because the percentage of forested areas has continually decreased

P. Thailand, because the percentage of forested areas has stayed the same

R. Philippines, because the percentage of forested areas suddenly decreased and then began to increase

S. Myanmar, because the percentage of forested areas has steadily decreased over time

T. Vietnam, because the percentage of forested areas has mostly increased

The table shows some common types of plastic debris cleaned from the ocean.

**Plastic Debris Items Removed from Oceans**

- Plastic straws ~ 3.6 million
- Plastic bottles ~ 1.5 million
- Plastic bottle caps ~ 1.4 million
- Plastic grocery bags ~ 1 million

Using the data in the table, which solution would **most** reduce the amount of plastic debris in the oceans?

A. switching from paper bags to plastic bags

B. recycling plastic bottles but not the cap

C. using paper straws instead of plastic straws

D. using larger plastic bags to carry groceries

## The questions on pages 9 - 10 refer to the passage(s) and image(s) shown.

### Lake Populations in Winter – Part 1

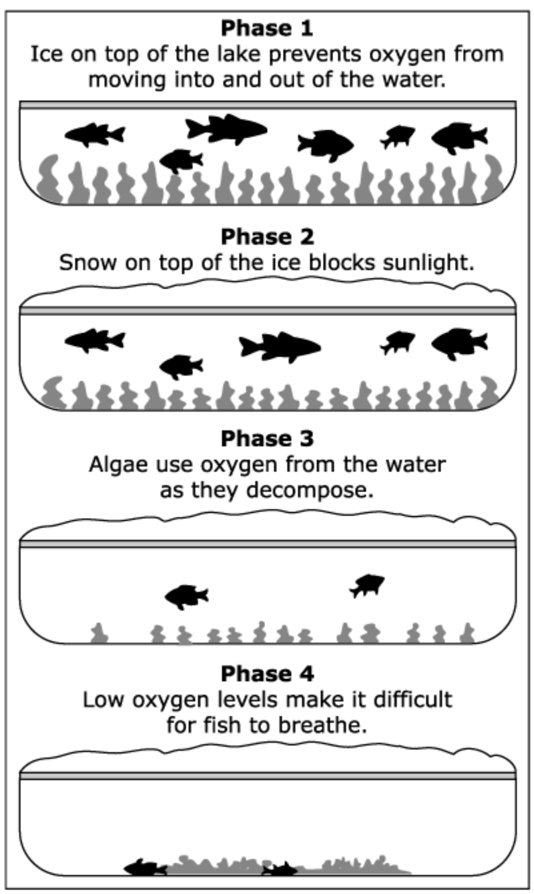
Scientists study population sizes of organisms living in a lake. The roles of these organisms are shown in Table 1.

**Table 1. Lake Organisms**

| Organisms | Role |
|---|---|
| Algae | Use water, sunlight, and carbon to produce oxygen |
| Zooplankton | Eat algae |
| Small fish | Eat zooplankton |
| Large fish | Eat small fish |

Scientists observe the lake for several years. They notice that each winter the surface of the lake freezes. In some years, the surface freezes for a short amount of time. In other years, a winter kill takes place. Figure 1 shows the events that take place to produce a winter kill.

**Figure 1. Winter Kill Stages**



## Lake Populations in Winter – Part 2

Table 2 shows data the scientists collected over several years for the populations of organisms in the lake. A winter kill took place in the winter between Year 1 and Year 2, in the weeks before the data for Year 2 were collected.

**Table 2. Population Data**

| Organism | Number of Organisms in Winter | | | | |
|---|---|---|---|---|---|
| | Year 1 | Year 2* | Year 3 | Year 4 | Year 5 |
| Algae | High | Very high | Medium | Low | Low |
| Zooplankton | Medium | Very high | Medium | Low | Very high |
| Small fish | Medium | Low | Medium | High | Medium |
| Large fish | High | Low | Low | Low | Medium |

*Year 2 data recorded just after winter kill

1. Based on the information in Part 1 and Part 2, how does a winter kill **most** affect an organism living in the lake?

   A. Less oxygen can be used by the large fish.

   B. More sunlight can be used by the algae to produce food.

   C. More decaying algae are available for zooplankton to eat.

   D. Less liquid water makes it more difficult for zooplankton to perform photosynthesis.

2. Based on the information in Table 1 (Part 1) and Table 2 (Part 2), which statement **best** explains the change in population size for some organisms from Year 4 to Year 5?

   M. As the small fish population increased, they ate much of the zooplankton, causing the amount of algae to decrease.

   P. As the large fish population increased, they ate some of the small fish, causing the amount of zooplankton to increase.

   R. As the algae population decreased, more food was available to decomposers, causing the large fish population to increase.

   S. As the zooplankton population decreased, less food was available to consumers, causing the small fish population to decrease.

3. A student claims that the population of large fish was harmed more than the population of small fish in the years after a winter kill.

   Which statement correctly describes evidence that **best** supports this claim?

   A. The large fish population size dropped greatly in the month following the winter kill.

   B. By Year 5, the population size of large fish was similar to the population size for small fish.

   C. By Year 4, the large fish population was small enough to be supported by the small fish population.

   D. The large fish population took a longer time to return to medium population size than the small fish population.

4. Although the lake studied by the scientists froze every year, a winter kill only took place in Year 2. Which statement **best** explains why a winter kill only took place during Year 2 of the study?

M. The snow covering the ice remaining on the lake prevented plants from producing oxygen, which is needed by small fish and large fish.

P. The larger amount of decay in the lake increased the amount of oxygen moving into the lake, causing much fewer algae to remain in the ecosystem.

R. The long amount of time that the lake was frozen caused much more oxygen to enter the water than in other years, causing algae to grow more slowly.

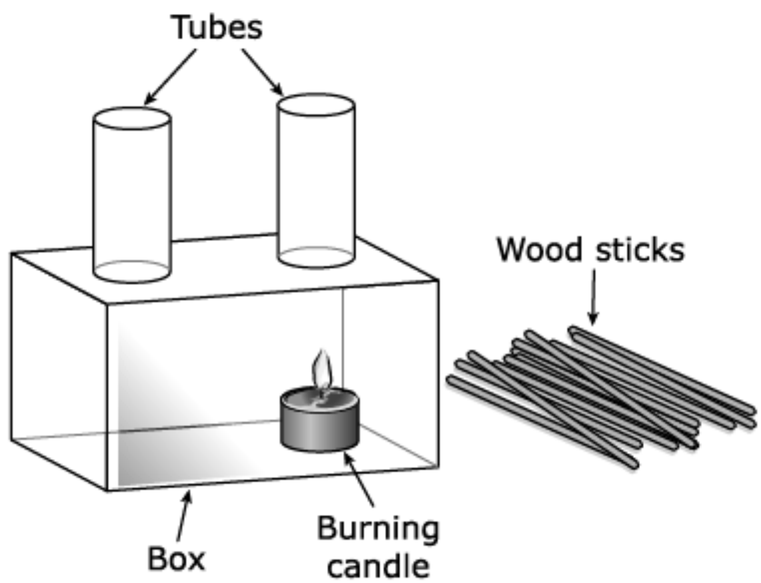
S. The lower levels of oxygen in the lake made it easier for zooplankton to use algae as a food source and more difficult for small fish to use zooplankton as a food source.

## The questions on pages 13 - 14 refer to the passage(s) and image(s) shown.

### Moving Curtain – Part 1

Students observe that the bottom of the curtain covering a classroom window moves out of the window when the teacher opens the window. They are surprised that the curtain moved because there is no wind outside the window. The students decide to investigate what caused the curtain to move toward the outside of the window. The students use a box with glass sides and attach two glass tubes to two holes in the top of the box. They place a lit candle underneath one of the tubes. Figure 1 shows the students' setup.

**Figure 1. Box and Candle Investigation Setup**

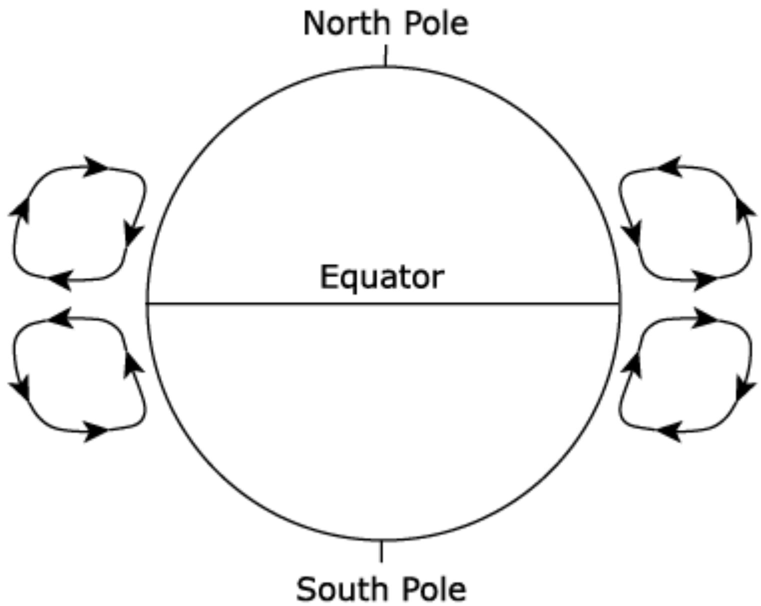



The teacher tells the students that each time they light a wood stick and hold the stick above the tube that is farthest from the burning candle, smoke from the burning stick will show the direction in which air moves inside the box. The students observe that the smoke from the wooden stick travels down into the tube and through the box.

## Moving Curtain – Part 2

The students learn that winds on Earth are created in a similar way. They find a model that shows wind patterns near Earth's equator. The model is shown in Figure 2. The arrows show the directions of the moving air.

**Figure 2.**
**Winds Near the Equator**



1. The students claim they can use the setup in Figure 1 (Part 1) to collect evidence that thermal energy is being transferred from the burning candle to the air.

   Which phrase describes evidence that would best support the students' claim if it were observed during the investigation?

   A. a decrease in the length of the candle

   B. an increase in the amount of air inside the box

   C. an increase in air temperature inside the tubes

   D. a decrease in the size of the flame above the candle

2. What is the source of energy that causes the wind patterns shown in Figure 2 (Part 2) to form?

   M. radiation from the sun

   P. convection from the sun

   R. radiation from Earth's interior

   S. conduction from Earth's interior

3. One student claims that the surface at the equator has a higher average temperature than the surface of areas to the north or south of the equator.

   Which statement **best** describes evidence from Figure 2 (Part 2) that supports the student's claim?

   A. Air that is near the equator rises.

   B. Air from the poles moves toward the equator.

   C. Air in the atmosphere moves toward the equator.

   D. Air that is near the equator sinks.

4. During the investigation, the students observed that the speed at which the smoke moved from the wood stick through the box was dependent upon the temperature difference between the ends of the box. When there was a bigger temperature difference, the smoke moved through the box with a faster speed. The students want to use the results of their investigation, and Figure 2 (Part 2), to explain why the bottom of the curtain moved out of the window when the teacher opened the window.

   Which explanation is best supported by their investigation and by Figure 2?

   M. The air outside was colder than the air inside, and air moved into the classroom.

   P. The air inside was colder than the air outside, and air moved out of the classroom.

   R. The air outside was warmer than the air inside, and air moved into the classroom.

   S. The air inside was warmer than the air outside, and air moved out of the classroom.

**Metadata – Science Items**

| Page Number | Item Type | Key | TN Standards |
|---|---|---|---|
| 1 | MS | P, R, S | 6.LS2.3 |
| 2 | MC | C | 6.LS2.5 |
| 3 | MC | M | 6.ESS3.2 |
| 4 | MC | D | 6.LS2.3 |
| 5 | MS | M, S | 6.ESS3.3 |
| 6 | MC | C | 6.ETS1.1 |

**Lake Populations in Winter**

| Page Number | Item Type | Item Number | Key | TN Standards |
|---|---|---|---|---|
| 7 | Stimulus | | | |
| 8 | Stimulus | | | |
| 9 | MC | 1 | A | 6.LS2.1 |
| 9 | MS | 2 | P | 6.LS2.1 |
| 9 | MC | 3 | D | 6.LS2.1 |
| 10 | MC | 4 | M | 6.LS2.1 |

**Moving Curtain**

| Page Number | Item Type | Item Number | Key | TN Standards |
|---|---|---|---|---|
| 11 | Stimulus | | | |
| 12 | Stimulus | | | |
| 13 | MC | 1 | C | 6.PS3.2 |
| 13 | MC | 2 | M | 6.ESS2.1 |
| 13 | MC | 3 | A | 6.ESS2.1 |
| 14 | MC | 4 | S | 6.PS3.2 |

| Metadata Definitions | |
|---|---|
| **Item Type** | MC = Multiple Choice<br>MS = Multiple Select<br>Stimulus = Cluster support information |
| **Key** | Correct Answer |
| **TN Standards** | Primary standard assessed in item. |